

NZLUSG 22/09/2026

Meeting 2 – NZLUM v0.4 2026 review and future direction: record of discussion, decisions and follow-up

MEETING / PROJECT	New Zealand Land-use Steering Group (NZLUSG) – Meeting 2: NZLUM v0.4 2026 review and future direction	DATE + TIME	22/09/2026, 14:00–15:55
LEAD	Richard Law (BSI)	MINUTE TAKER	James Ardo (BSI)
LOCATION / LINK	Virtual (Microsoft Teams); recording and transcript held by BSI	NEXT MEETING	Not scheduled. Follow-up by correspondence (change-proposal ranking, see Actions 1–4).

PURPOSE / DESIRED OUTCOME

Review the NZLUM v0.4 classification system and the NZLUM v0.4 2026 land use map (changes since v0.3 and validation results), and hold an open discussion of suggested changes for the next iteration. No change decisions were sought at the meeting: all suggestions are to be collated and ranked by members afterwards.

ATTENDEES

PRESENT

Anne-Gaelle Ausseil (MCERT), Michael Bates (MCERT), Deborah Burgess (MCERT), Greg Byrom (LINZ), Robert Cardwell (Market Economics), Ashton Eaves (HBRC), Haydon Jones (WRC), Richard Law (BSI, chair), James Ardo (BSI, minutes)

APOLOGIES / ABSENT

None recorded

AGENDA

#	ITEM	LEAD	TIME
1	Presentation: NZLUM v0.4 and the 2026 land use map – changes, validation, report recommendations and MCERT/BSI suggested changes	Richard Law	14:00 – 14:58
2	Open floor: questions, discussion of suggested changes, new suggestions, regional council requirements	All	14:58 – 15:49
3	Close and next steps	Richard Law	15:49 – 15:55

DISCUSSION + DECISIONS

Initials: RL Richard Law (BSI); JA James Ardo (BSI); GB Greg Byrom (LINZ); RC Robert Cardwell (Market Economics); AE Ashton Eaves (HBRC); HJ Haydon Jones (WRC); DB Deborah Burgess (MCERT); A-GA Anne-Gaelle Ausseil (MCERT); MB Michael Bates (MCERT). Items are in the order discussed. ACTION column refers to the Action Register.

#	RAISE D BY	NOTES / KEY POINTS	OUTCOME / DECISION	OWNER	ACTION
1	GB	LINZ is substantially improving the links between DVR records, titles and parcels. This does not change DVR attribute content (e.g. zone codes, granularity). LINZ has good contacts with council rating staff and can raise DVR data problems with them on BSI's behalf. RL: DVR zone codes are not comparable across regions (the same code means different things in different districts), which limits their use as evidence.	GB to look at how zone coding varies across councils and advise BSI. BSI to pass specific DVR data issues to LINZ for follow-up with councils.	GB / BSI	5

#	RAISED BY	NOTES / KEY POINTS	OUTCOME / DECISION	OWNER	ACTION
2	RC	Proposal: relabel or redefine residential tertiary classes 3.1.1–3.1.3 as a housing typology (standalone, terraced, apartment) rather than high/medium/low density. LINZ publishes a ~1 m digital surface model; building heights derived from it (or LiDAR) would be strong evidence for these classes and useful at parcel level for ME's modelling. RL: the report already recommends building heights; the poor 3.1.1/3.1.2 validation result was unexpected.	Change proposal (residential class labels/definitions) and data proposal (building heights) to go into the change-proposal document for ranking.	BSI	1
3	RC	Street-level imagery (Google Street View or open equivalents) could support validation in urban areas: looking sideways rather than down. Vision-language models reading signage could also help distinguish retail, hospitality and office uses within 3.3.0 Commercial.	Consider street-level imagery in the validation tooling, and as a future evidence source for commercial subtypes. To change-proposal document.	BSI	1
4	AE	Will H3 resolution 12/13 be computationally limiting? HBRC could not process the Hawke's Bay region at resolution 12 (≈ 49 M cells vs 7 M at resolution 11). RL: the compacted DGGS representation now used in LUIS makes higher resolution tractable. The published product is dissolved, so growth is at feature edges only, not $7\times$ more data. Same schema at higher fidelity. Resolution 12 is $\approx 7\times$ finer nationally; resolution 13 $\approx 49\times$ finer in urban areas. Coarse resolution-11 cells spanning two properties are a likely contributor to poor urban (Class 3) validation. A property-boundary product is complementary, not a replacement.	Resolution increase remains a proposal for ranking; output size implications noted for users.	BSI	1
5	AE	Has alignment with ANZSIC (as used by ABARES) progressed, so that land-use outputs can be linked to statistical/economic data? RL: not pursued to date. AE: linking cropping and horticulture classes to sector output (e.g. regional squash or carrot production) would provide a plausibility check on mapped areas.	Fold into the proposed validation of mapped areas against APS and sector totals. ANZSIC concordance parked.	BSI	1
6	RC	Census population totals (SA1) could be used to constrain or validate residential (and commercial) classes. RL: a population-density attribute could be derived in the same way as the proposed stocking-rate attribute; useful for density tiers and for identifying crowding.	Data/attribute proposal (population density) to change-proposal document.	BSI	1
7	DB / GB	DB: the Stats NZ–LINZ property spine work aimed to put people in buildings (residential vs non-residential footprints); worth connecting with. GB: the property spine pilot closed about two years ago; the work continues as a dwellings register for the next census, at an early stage (contact: Mike Judd, LINZ, formerly seconded to Stats NZ). A separate LINZ building spine project is under way; prototype data is currently only shared with NHC but is intended to be made more widely available (project lead: Susan Shaw). DB: demand for this is growing, e.g. climate adaptation (which people and assets are at risk).	Potential inputs for Class 3 and for a property-scale product. BSI to follow up via GB.	GB / BSI	7

#	RAISED BY	NOTES / KEY POINTS	OUTCOME / DECISION	OWNER	ACTION
8	AE	AE intends to integrate the NZLUM v0.4 classification within the next couple of months; how much is it likely to change? RL: depends on the ranking outcome. There is still freedom to make necessary changes now ("be bold now"); any large departure will be flagged in the change-proposal document so users can object. Caution about change will increase once the classification settles and validation is acceptable. Kart changesets of input data will allow method changes (as distinct from classification changes) to be replayed over earlier inputs.	No freeze committed. Major departures from v0.4 to be flagged explicitly in the change-proposal document.	RL	1
9	AE	Could the classification be applied backwards to e.g. 2020? RL: not feasible; key inputs (e.g. dairy consents) have no accessible history before BSI began archiving weekly changes; only some inputs have historical versions. AE: trends will therefore build forward from 2025/2026; adopters should expect a method-driven step change on adoption.	Noted. Distinguishing real change from method change remains an open issue (Meeting 1 item I).	—	—
10	RC	Could users with the time and skills rebuild their own variant of the map from the inputs and rules? RL: the classification rules and input configuration are open on GitHub; the LUIS software is not open (whether it should be is an open question). BSI is experimenting with embeddings (AlphaEarth-style, 10 m, 64-dimension vectors) from which many classifications could be derived. RL: NZLUM's distinct value is an agreed classification with a validated accuracy figure. DB: the framework was always designed to allow other classifications beneath it. A-GA: the aim is one reasonably authoritative 80:20 classification plus a one-stop catalogue of inputs and a stack of derived products. RL: the status quo is roughly half of regions with incompatible bespoke maps; a consistent schema lets a model built for one region be applied in another. RC: much of the value is in the assembled inputs, not only the final map.	No change to approach. Open-sourcing LUIS parked for a future meeting.	—	—
11	RL	Which unmapped tertiary classes would be most valuable? RL: horticulture and cropping subclasses are the obvious candidates but current data cannot support them (see report recommendation 4).	To be answered through the ranking document.	All	3
12	A-GA	MCERT now spans cities, environment, regions and transport: are there Ministry of Transport, KiwiRail or utility datasets that could be added as inputs? RL: more corroborating evidence is always useful; agreement between independent sources (e.g. KiwiRail and LINZ on rail assets) raises confidence.	A-GA to identify candidate transport/utility datasets and contacts within MCERT's new scope.	A-GA	8
13	GB	LINZ is building a dataset of land without a title (a prototype is on LDS, labelled pilot). Statutory actions on parcels record purpose (e.g. KiwiRail land; historic set-asides), which has not been exploited before. LINZ is trialling AI to classify statutory-action purposes, with promising early results; many actions are not digitised and are unlikely to be. Pilot publication expected in 6–12 months. RL: also valuable for PAN-NZ (protected areas) where councils lack registers.	BSI to track as a future input. GB to keep the group informed.	GB	7

#	RAISED BY	NOTES / KEY POINTS	OUTCOME / DECISION	OWNER	ACTION
14	RL / AE / HJ	RL asked what regional councils need from the map that they cannot yet do. AE: direction is right. Trend analysis is the core need, especially explaining drivers of water-quality change; needs several years of consistent data. HJ: more subdivision of food production types and intensity of use, particularly cropping and horticulture; intensity relates to water quality and other environmental effects. AE/HJ: highly productive land (NPS-HPL; LUC classes 1–3) and its fragmentation and loss to urban use; HBRC runs a separate parcel-change workflow that NZLUM change tracking could support. Soils (S-map) and LUC as joinable attributes would help.	Priorities noted for ranking. Attribute proposal: LUC/soils join.	BSI	1
15	DB / GB / RL	DB: the dissolved DGGS product cannot show fragmentation because ownership boundaries are invisible; getting 3.1.4 and 2.9.0 right and offering a property-based product addresses this. Property vs parcel: GB: cadastral parcels are survey units and not meaningful for use; property boundaries are closer to what is needed and change less often. DB: property boundaries shift over time and rely on the improving LINZ property dataset. GB: a review of the Rating Valuation Rules is under way and may deliver more granular property-level data; details not yet known.	DECISION: the proposed aggregated product will use property boundaries, not cadastral parcels. The hexagon (DGGS) product remains the primary output.	BSI / GB	6
16	RL / GB / AE / DB	How should a property with several uses be recorded (e.g. a sheep and beef farm that is 60% forestry by area; Cape Kidnappers, ≈1,000 ha coded as accommodation)? GB: DVR apportionments split value across uses within a property but are not attached to titles or land areas; whether floor area is recorded per apportionment is to be checked. DVR is valuation data applied to another purpose, but it is the best available. AE: S-map records proportions (dominant plus percentages); a similar approach could work. DB: an AgriBase-style list of the top 3–4 uses with percentage cover per property would support policy analysis (e.g. how many properties have a qualifying area of sheep and beef). RL: the hexagon product will still be published, so users can aggregate differently if they disagree.	Proposal for the property product schema: dominant use by area plus the next 3–4 uses with area share, retaining commodities. To change-proposal document.	BSI / GB	1, 6
17	RC / RL / AE	RC: vertical mixed use (ground-floor shops with apartments or offices above) is common on main streets and matters for transport and economic modelling; RC currently assigns it a class of its own. RL: NZLUM assumes one dominant use in two dimensions, a blind spot: small by national area (<1%) but significant for urban modelling; unclear what to record (ground floor vs stack) or how to validate it. 3D representation was also raised at Meeting 1. Rural analogue: farm stays and other co-located uses. RC: spatial precision matters less than recording co-location so that totals add up. Suggest consulting HBA and capacity-for-growth practitioners. AE offered to connect RC and RL with Auckland Council spatial planners who build 3D capacity models.	Logged as an unresolved issue; suggestions invited through the ranking form. AE to make introductions.	AE / RC / RL	9

#	RAISED BY	NOTES / KEY POINTS	OUTCOME / DECISION	OWNER	ACTION
18	RL / MB	RL and JA to compile all suggestions (addendum report, MCERT/BSI list, Meeting 1 carry-overs, today's discussion) into a document for members to rank and comment on, with room for new ideas. Feedback wanted within weeks to a month. No contract for the 2027 map yet: the meeting informs a BSI proposal to MCERT (MB: BSI's timing is fine). Both organisations are restructuring, but the priority list will stand regardless. No further NZLUSG meeting planned in the near term.	DECISION: no change decisions taken at the meeting, by design; prioritisation via written ranking. Presentation to be shared.	RL / JA	1-4

FOLLOW-UP

ACTION REGISTER

#	ACTION	OWNER	DUE	STATUS
1	Circulate a change-proposal document collating all suggestions (report recommendations, MCERT/BSI list, Meeting 1 carry-overs, this meeting) for comment and ranking; flag any major departures from v0.4.	RL / JA	25/09/2026	OPEN
2	Upload presentation, minutes and related materials to nzlum.landcareresearch.co.nz .	RL	25/09/2026	OPEN
3	Return rankings and comments on the change-proposal document, including any new suggestions (e.g. on recording mixed use).	All members	24/10/2026	OPEN
4	Collate responses, agree priorities for the 2027 map, and prepare the BSI proposal to MCERT.	BSI / MCERT	31/10/2026	OPEN
5	Review how DVR zone coding varies across councils and advise BSI; act as conduit for DVR data issues to council rating staff.	GB	31/10/2026	OPEN
6	Advise whether DVR apportionment records carry floor area or land area, and what the Rating Valuation Rules review will add at property level.	GB	31/10/2026	OPEN
7	Provide contacts and details for the untitled-land / statutory-actions prototype, the building spine (Susan Shaw) and the dwellings register (Mike Judd).	GB	31/10/2026	OPEN
8	Identify Ministry of Transport, KiwiRail and utility datasets (and contacts) available through MCERT that could be added as inputs.	A-GA	31/10/2026	OPEN
9	Introduce RC and RL to Auckland Council spatial-planning contacts (3D capacity modelling) regarding mixed-use representation.	AE	31/10/2026	OPEN
10	Contact the top LRIS downloaders of the NZLUM layers for feedback on use and improvements (carried over from Meeting 1, action N).	RL / JA	30/11/2026	OPEN
11	Assess feasibility of aerial-imagery segmentation for industrial land (with Lincoln University 'AI for Land Use' placement).	JA	31/03/2027	IN PROGRESS

PARKING LOT / LATER

ITEM	WHEN / WHERE TO REVISIT
Recording multiple and vertically stacked uses (3D / mixed use) and how to validate them	Change-proposal document; revisit when the property-product schema is drafted
Whether the LUIS classifier software should be open-sourced	Next NZLUSG meeting
ANZSIC concordance for linking land use to economic statistics	Alongside validation against APS / sector totals
Distinguishing real land-use change from method change (Kart replay of inputs)	When the 2027 map is produced
Carried over from Meeting 1: defence land under Class 3; placement of 3.2.3 Community services; tertiary subdivision of 2.9.0; formal standard (ISO 19144-3 LUML) and ABARES contact	Via change-proposal document; standards item unscheduled

AT-A-GLANCE CLOSE

KEY DECISION(S)	1. No changes to the classification system or map were decided at the meeting, by design; all suggestions will be ranked by members through a written change-proposal document (Actions 1–4). 2. The proposed aggregated product will use property boundaries rather than cadastral parcels; the hexagon (DGGS) product remains the primary output. 3. Feedback window of about one month; BSI to carry agreed priorities into its proposal to MCERT.
MEETING CLOSED	15:55
MINUTES APPROVED BY	James Ardo (draft); to be confirmed by Richard Law